

Pedro Oliveira

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Profile

Mathematics undergraduate specialising in statistical modelling, machine learning and computational methods. Experienced in applying quantitative techniques to real-world datasets through academic and independent projects using Python, R and SQL.

Education

University of Sheffield

- BSc IN MATHEMATICS (GRADUATING IN JULY 2026) September 2023 - July 2026
- Expected classification: First Class (or 2:1)
 - Relevant modules: Machine Learning, Operations Research, Time Series, Stochastic Processes and Finance, Statistical Inference and Modelling
 - Focus areas: Statistical modelling, data analysis, machine learning, optimisation and computational methods

Technical Skills

Programming Languages	Python (Pandas, NumPy, Matplotlib, SciPy), SQL, R (tidyverse), C++, C
Tools	Excel, Git, Github, Visual Studio
Spoken Languages	English, Portuguese

Projects

- Red vs Grey Squirrel Population Analysis 🌐 pedroddvo/GreyVsRed
- Collected and cleaned UK wildlife observation data from NBN Atlas
 - Studied relationships between grey squirrel spread and red squirrel population trends
 - Performed exploratory data analysis and statistical modelling
 - Produced visualisations highlighting regional and temporal trends
 - Managed project development using Git and published code and documentation on GitHub

Academic Projects

- Statistical Modelling of Paper Helicopter Dynamics (1st Class) 🌐 pedroddvo/PaperHelicopter
- Designed and conducted a statistically rigorous experiment investigating factors affecting paper helicopter flight performance
 - Analysed data from 150+ experimental flights using R, exploratory data analysis and visualisation techniques
 - Applied design of experiments methodology to evaluate the effects of multiple experimental variables
 - Developed and interpreted linear models to quantify relationships between design parameters and flight characteristics
 - Produced a formal technical report using LaTeX, documenting methodology, statistical analysis and conclusions
 - Created supporting visualisations and 3D models to communicate experimental design
- Fast Fourier Transform and Signal Analysis (1st Class) 🌐 pedroddvo/FFTPresentation
- Delivered a technical presentation explaining FFT algorithms and their applications in signal processing
 - Explored the computational advantages of FFT algorithms compared with direct discrete Fourier transform methods
 - Implemented FFT algorithms to analyse frequency components within sample datasets
 - Communicated complex mathematical concepts to a technical audience through a formal presentation

Interests

Technical	Data Science, Mathematical Modelling, Open-source, Linux
Personal	Music production, Piano, Golf

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